CS-4982 · FINAL REPORT

High-Stakes Hedging Detection

A Transductive Approach to Predict Company Failure from Annual Reports using NLP & Machine Learning

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SYNTHETIC MODE + SEDAR+ READY

7 Pipeline Stages	**HI** Composite Score
21 APA References	**~5Q** Lead Time Signal

01 Project Goals

What this project set out to achieve

A Core Problem

Detect hidden patterns of selective reporting in financial disclosures. Companies emphasise positive metrics while obscuring negatives, misleading investors and regulators.

B Hedging Index

Construct a novel composite score (HI) combining NLP linguistic signals and quantitative financial signals to measure informational opacity in a single reproducible metric.

C Predictive Model

Train ML classifiers (Logistic Regression, Random Forest, XGBoost) to estimate the probability of near-term company failure using the HI and its three sub-components.

D Stakeholder Output

Produce actionable early-warning outputs — ranked watchlists, per-company risk scorecards, and a 5-panel visualisation dashboard accessible to non-technical audiences.

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Key Concepts

The three pillars underpinning the Hedging Detection framework

01 Financial Hedging

The Problem Domain

- Companies use MD&A language to overemphasise positive metrics while burying negatives in dense qualificatory prose
- Traditional auditing detects numerical errors but misses rhetorical obfuscation and selective emphasis
- Linguistic hedging signals precede formal distress indicators by approximately 5 quarters

02 Natural Language Processing

The Linguistic Signal

- Loughran-McDonald (2024) dictionary: 33 hedging terms across 7 semantic categories
- Doc2Vec (Le & Mikolov, 2014): 300-dimensional document embeddings capturing distributional meaning
- FinBERT (Yang et al., 2020): contextualised sentiment scoring — planned next phase

03 Hedging Index (HI)

The Core Contribution

- $HI = 0.40 \times LHS + 0.35 \times FOS + 0.25 \times VS$ — composite score on [0,1]
- Integrates linguistic opacity, financial overemphasis, and ratio volatility into one metric
- Risk tiers: Low <0.25 | Medium <0.45 | High <0.65 | Critical ≥0.65

03 Solution Architecture

7-layer modular pipeline from raw data ingestion to stakeholder output

1	2	3	4	5	6	7
Data Ingestion	**Pre-processing**	**Feature Eng.**	**Hedging Index**	**ML Modelling**	**Valida-tion**	**Report-ing**
EDGAR SEDAR+ APIs	Clean Tokenise Align	LHS FOS VS + NLP	HI=0.40 LHS +0.35 FOS +0.25 VS CORE	LR · RF XGBoost SHAP	Backtest Sensitivity	Dashboard Scorecards CSV

STAKEHOLDER OUTPUTS

- Risk Scorecard
- Ranked Watchlist
- 5-Panel Dashboard
- SHAP Explanations
- CSV Export

Modular design — each stage is independently testable, replaceable, and extensible.

04 Hedging Index — Core Contribution

The composite distress indicator at the heart of this research

$$HI = (0.40 \times LHS) + (0.35 \times FOS) + (0.25 \times VS)$$

LHS — 40%
Linguistic Hedging Score

Density of Loughran-McDonald hedging/uncertainty terms per 100 words of MD&A text. Normalised to [0,1]. Higher = more opaque language.

0.181 → 0.741

Healthy → Failed

FOS — 35%
Financial Overemphasis Score

Company P/E ratio deviation from sector median. Positive deviation signals selective metric emphasis while obscuring weaker fundamentals.

0.142 → 0.563

Healthy → Failed

VS — 25%
Volatility Score

Rolling 2-quarter standard deviation of Debt/Equity and Current Ratio, normalised by historical mean. Proxy for financial instability.

0.098 → 0.412

Healthy → Failed

05 Datasets & Key Resources

Data sources, frameworks, and libraries underpinning the implementation

DATA SOURCES

1 SEC EDGAR

10-K / 10-Q structured filings
XBRL-tagged financial fields
Full-text MD&A sections

2 EDGAR-Corpus

Loukas et al. (2021)
Billions of tokens
NLP training dataset

3 SEDAR+

sedarplus.ca
Canadian company filings
PDF / HTML reports

4 LM Dictionary

Loughran & McDonald (2024)
33 hedging terms
Financial-domain lexicon

FRAMEWORKS & LIBRARIES

Pandas / NumPy

Data manipulation & cleaning

NLTK / spaCy

NLP preprocessing pipeline

Gensim Doc2Vec

300-dim document embeddings

scikit-learn

ML classifiers & evaluation

XGBoost / SHAP

Gradient boosting + explainability

Matplotlib/Seaborn

5-panel visualisation dashboard

06 Success Criteria & Metrics

How we define and measure project success

SUCCESS CRITERIA

A Accurate prediction of near-term company distress using the Hedging Index and ML classifiers (LR, RF, XGBoost)

B Statistically significant hedging behaviours distinguishing healthy from distressed company profiles

C Robustness of the HI framework across sectors — Hedge Funds, Quant Funds, and Mutual Funds

D Interpretable, stakeholder-accessible outputs requiring no specialist quantitative expertise

VALIDATION METRICS

ROC-AUC	Discriminative power of the failure classifier
Precision	Proportion of flagged companies truly at risk
Recall + F1	Coverage of actual failure events detected
HI Sensitivity	Stability of risk tiers under weight variation
Lead Time (~5Q)	Quarters ahead of failure the HI signal emerges

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Key Results

Simulation outcomes across 8 companies × 12 quarters

100%
Tier classification accuracy (simulation)

~5Q
Lead time before failure event

92.4%
Highest failure probability flagged

0.598
Mean HI — failed company profile

RISK REPORT — COMPANIES RANKED BY FAILURE PROBABILITY

Ticker	Company	Avg HI	Fail Prob.	Risk Tier
FLUX	Flux Money Mgmt	0.700	92.4%	CRITICAL
GRVY	Gravity Partners	0.577	78.0%	HIGH
ALPH	Alpha Capital	0.568	76.5%	HIGH
ECHO	Echo Global Fund	0.529	68.7%	HIGH
DELT	Delta Quantitative	0.444	48.4%	MEDIUM
CRUX	Crux Investments	0.404	38.7%	MEDIUM

Note: Counterintuitive results (healthy companies ranked above failed) are an artefact of synthetic FOS data — discussed in paper Section 5.1.3

08 Relevance to Stakeholders

Who benefits from this framework and how

PRIMARY

Investors (Institutional & Retail)

TOOL: Ranked watchlist + failure probability scorecard

Rebalance portfolios before distress announcements. Linguistic signal emerges ~5 quarters ahead of failure — sufficient time for meaningful action.

PRIMARY

Regulators (SEC / IIROC)

TOOL: HI heatmap across all companies × quarters

Target oversight resources toward companies with sustained or escalating hedging patterns. Particularly valuable in resource-constrained regulatory environments.

PRIMARY

Auditors & Forensic Accountants

TOOL: Component breakdown — LHS / FOS / VS diagnosis

Identify which dimension of hedging is driving elevated risk — linguistic, financial, or volatility — and direct forensic investigation accordingly.

SECONDARY

Creditors & Capital Markets

TOOL: Failure prob. vs. D/E and ROE scatter plots

Supplement traditional credit analysis with a behavioural dimension absent from ratio models. Assess counterparty risk beyond reported financial figures.

09 Project Contributions

Three major advances across computer finance, NLP, and corporate governance

01 Hedging Index Metric

Creates a novel composite metric combining language and quantitative signals to measure informational opacity. Unlike existing workable metrics, the HI evaluates linguistic and financial signals jointly, giving a holistic view of how companies obscure information.

02 Linguistic Lead Indicators

Demonstrates that linguistic signals (LHS via Loughran-McDonald dictionary) precede formal financial distress by approximately 5 quarters. This provides earlier warning than traditional numerical models (Altman 1968, Beaver 1966) which rely on accounting ratios alone.

03 Open, Modular Architecture

Delivers a reproducible, extensible pipeline where each component is independently testable and replaceable. Transparent weight calibration and documented design choices enable peer critique and iterative improvement in future research phases.

10 Next Steps & Future Work

Priority extensions for the next development phase

1 Real EDGAR / SEDAR+ Data

Migrate from synthetic simulation to live 10-K filings from EDGAR and Canadian PDF reports from SEDAR+. Validate the HI on companies that have formally failed or been acquired.

2 FinBERT Integration

Replace Doc2Vec embeddings with FinBERT contextualised sentence representations to capture subtle hedging language that bag-of-words approaches miss in real financial prose.

3 Cross-Industry Calibration

Extend the HI beyond financial firms (Hedge/Quant/Mutual Funds) to broader sectors. Adjust sector-specific FOS thresholds to account for industry-specific P/E norms.

4 Network Contagion Analysis

Add inter-firm network layer using NetworkX (Hagberg et al., 2008) to detect coordinated hedging across industry peers — a systemic risk signal absent from single-company analysis.

5 Public-Facing Web Dashboard

Build an accessible web interface allowing non-technical users to filter results by industry, time period, and risk tier — democratising access to early-warning financial intelligence.

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Conclusion

Summary of findings and broader significance

1. Presents a framework for detecting financial hedging behaviour in corporate disclosures and translating it into early-warning signals of near-term company failure.

2. The Hedging Index (HI) successfully stratifies companies by financial health, with the LHS component providing a ~5-quarter lead time over conventional distress indicators.

3. The 7-stage pipeline is modular, reproducible, and extensible — each component is independently testable and documented for peer critique.

4. Limitations are acknowledged: synthetic data, preliminary weight calibration, and counterintuitive FOS artefacts. A clear road-map addresses each in future phases.

5. Broader significance: the gap between what companies disclose and what is true is not merely qualitative — it is a quantifiable signal that modern NLP methods are well-positioned to surface at scale.

12 References & Resources

21 APA references — key sources hyperlinked

Loughran & McDonald (2024)
LM Master Dictionary

Loukas et al. (2021)
EDGAR-Corpus dataset

Altman (1968)
Discriminant analysis & bankruptcy

Beaver (1966)
Financial ratios as predictors

Le & Mikolov (2014)
Doc2Vec — distributed representations

Yang et al. (2020)
FinBERT pretrained LM

Devlin et al. (2019)
BERT — deep bidirectional transformers

Chen & Guestrin (2016)
XGBoost scalable boosting

Lundberg & Lee (2017)
SHAP — unified model interpretation

Řehůřek (2023)
Gensim Doc2Vec tutorial

Pedregosa et al. (2011)
scikit-learn

Harris et al. (2020)
NumPy array programming

Hunter (2007)
Matplotlib 2D graphics

McKinney (2010)
Pandas data structures

Hagberg et al. (2008)
NetworkX

Waskom (2021)
Seaborn statistical viz

SEC EDGAR (2024)
EDGAR full-text search

SEDAR+
Canadian company filings

Mikolov et al. (2013)
Word2Vec — distributed word representations

Abney (2007)
Semisupervised learning for CL

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